

Scalability & Future Plan

Date	15 July 2026
Team ID	SWUID2026217449
Project Name	Rising Waters - Flood Prediction ML
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses a synthetically generated dataset rather than real-world data	May not fully capture real regional weather patterns	Medium
2	Single-region model without geographic specificity	Reduces accuracy for location-specific predictions	Medium
3	No user authentication or historical prediction storage	Limits personalization and long-term tracking	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handles single-user local Flask requests	Deploy on IBM Cloud with load balancing for concurrent users
2	Data Storage	Static local CSV dataset	Integrate real-time weather API and cloud database for live data
3	Performance	Model loaded fresh per request	Cache model in memory and serve via Unicorn/WSGI for faster inference
4	Security	No authentication or input validation layer	Add user login, HTTPS, and stricter input validation

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Integrate real-time weather API data (e.g., OpenWeatherMap)	1 Month	Live, continuously updated flood predictions
Phase 3	Add region-specific models for better local accuracy	2 Months	More precise, location-aware flood risk assessment
Phase 4	Cloud deployment on IBM Cloud with alert notifications	3 Months	Scalable platform with SMS/email flood alerts